Advanced · AI systems vocabulary · Highlights · IBM Technology

Seven Terms That Describe a Modern AI System

Place agents, reasoning models, vector search, RAG, MCP, mixture of experts and ASI on one map, and see how they connect.

For engineers who keep meeting these terms and want the mechanism behind each, not the marketing · 26 July 2026

The one- or two-page version: what matters, the topics it covers, and every term defined. Read the full lesson for the explanations and worked examples.

Watch the source video · Read the full lesson · Read the condensed version

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What matters

- ✓ An agent is a loop — perceive, reason, act, observe — around a model that is otherwise stateless and reactive.
- ✓ Large reasoning models are fine-tuned to work step by step, trained on problems whose answers can be checked automatically.
- ✓ A vector database stores embeddings so that 'find similar' becomes a distance computation rather than a keyword match.
- ✓ RAG uses that search to inject relevant text into the prompt, giving a model facts it was never trained on.
- ✓ MCP standardises how applications expose tools and data to models, replacing one-off connectors with a common protocol.
- ✓ Mixture of experts activates only a few subnetworks per token, so total parameters and active parameters come apart.
- ✓ MoE saves computation, not memory — every expert still has to be resident even though few are used.
- ✓ AGI and ASI are both theoretical; ASI additionally supposes recursive self-improvement, which is what makes it a different claim.

What it covers

01 **Agentic AI** `0:40`

Systems that reason and act autonomously toward a goal, cycling through perceive, reason, act and observe instead of answering once.

02 **Large reasoning models** `1:23`

LLMs given reasoning-focused fine-tuning so they work through a problem step by step before answering.

03 **Vector databases** `2:46`

Data stored as embeddings rather than raw blobs, so similarity search becomes a mathematical operation.

04 **Retrieval augmented generation** `4:09`

Use vector search to pull relevant passages and paste them into the prompt, so the model answers from facts it was never trained on.

05 **Model Context Protocol** `5:33`

A standard way for applications to expose tools and data to models, replacing bespoke connectors with one protocol.

06 **Mixture of experts** `6:15`

Split the model into specialised subnetworks and activate only the few each token needs, so size and cost come apart.

07 **AGI and ASI** `8:17`

Two theoretical thresholds. AGI matches human experts across cognitive tasks; ASI goes beyond, with recursive self-improvement.

Glossary

Agentic AI — A system that pursues a goal autonomously by cycling through perceiving, reasoning, acting and observing, rather than answering a single prompt.

Chain of thought — The intermediate reasoning a model generates before its final answer. What a chatbot is producing while it shows that it is thinking.

Vector database — Storage built to hold embeddings and retrieve them by similarity, turning 'find things like this' into a distance computation.

Silent failure — When retrieval misses the passage containing the answer. No error is raised — the model simply answers from the wrong material, fluently.

MCP server — The component fronting an external system, declaring what a model can reach there and how. The right place to enforce permission.

Mixture of experts (MoE) — An architecture splitting a model into specialised subnetworks, with a router activating only a few per token.

Active parameters — The fraction of a MoE model's parameters used for a given token. Determines compute cost, while total parameters determine memory.

ASI — Artificial superintelligence: theoretical capability beyond human level, potentially able to redesign and improve itself recursively.

Large reasoning model — An LLM fine-tuned to work through problems step by step before answering, trained on tasks whose answers can be checked automatically.

Embedding model — A model that converts text, images or audio into a vector whose position encodes the content's meaning.

RAG — Retrieval augmented generation: searching a corpus for relevant passages and inserting them into the prompt so the model can answer from them.

MCP — Model Context Protocol: a standard for applications to expose tools, data and prompt templates to models, replacing per-pairing connectors.

Tool vs resource — In MCP, a tool is an action the model invokes; a resource is data the client reads. Something to do versus something to read.

Router / gating network — The small learned layer that scores experts for each token and selects the top few to run.

AGI — Artificial general intelligence: a theoretical system able to complete all cognitive tasks as well as any human expert.